

Frontend Interview Questions and Answers (0–2 Years Experience)

Topics Covered:

- HTML
 - CSS
 - JavaScript
 - TypeScript
-

HTML Interview Questions

1. What is HTML?

Answer

HTML stands for HyperText Markup Language. It is used to structure web pages using elements and tags such as headings, paragraphs, tables, forms, and images. HTML forms the foundation of every webpage.

2. What are semantic tags?

Answer

Semantic tags clearly describe the meaning of the content inside them. They improve readability, SEO, and accessibility. Examples include header, footer, article, section, and nav.

3. Difference between div and span?

Answer

A div is a block-level element that takes full width and starts on a new line. A span is an inline element that only takes required width and stays within the same line. Div is mainly used for layouts, while span is used for styling small text portions.

4. What is the difference between id and class?

Answer

An id uniquely identifies a single HTML element, while a class can be reused across multiple elements. IDs are accessed using # in CSS, and classes are accessed using dot notation. Classes are commonly used for reusable styling.

5. What are HTML forms?

Answer

HTML forms are used to collect user input such as text, passwords, emails, and selections. Forms contain elements like input, textarea, select, checkbox, and button. They are commonly used for login, registration, and search functionality.

6. Difference between GET and POST?

Answer

GET sends data through the URL and is mainly used for fetching data. POST sends data in the request body and is used for secure data submission such as login or registration. POST is more secure than GET for sensitive information.

7. What are meta tags?

Answer

Meta tags provide metadata about the webpage, such as description, keywords, charset, and viewport settings. They help browsers and search engines understand the page better. Meta tags are placed inside the head section.

8. What is localStorage?

Answer

localStorage is a browser storage mechanism used to store data permanently in the browser. The stored data remains even after closing and reopening the browser. It is commonly used for storing preferences and tokens.

9. What is sessionStorage?

Answer

sessionStorage stores data only for the current browser session. Once the browser tab or window is closed, the stored data is removed. It is useful for temporary session-related data.

10. Difference between localStorage and sessionStorage?

Answer

localStorage keeps data permanently until manually cleared, while sessionStorage clears data after the browser session ends. Both store data as key-value pairs in the browser. localStorage is commonly used for persistent settings.

CSS Interview Questions

11. What is CSS?

Answer

CSS stands for Cascading Style Sheets and is used to style HTML elements. It controls colors, layouts, spacing, fonts, and responsiveness. CSS helps create visually attractive web applications.

12. What is specificity?

Answer

Specificity determines which CSS rule is applied when multiple rules target the same element. Inline styles have highest priority, followed by IDs, classes, and element selectors. Higher specificity overrides lower specificity.

13. Difference between inline, internal, and external CSS?

Answer

Inline CSS is written directly inside HTML elements using the style attribute. Internal CSS is written inside the style tag within the HTML file. External CSS is written in separate .css files and is preferred for maintainability.

14. What is Flexbox?

Answer

Flexbox is a one-dimensional CSS layout system used to align and distribute items efficiently. It simplifies horizontal and vertical alignment. Flexbox is commonly used for navigation bars, cards, and responsive layouts.

15. What is CSS Grid?

Answer

CSS Grid is a two-dimensional layout system that handles both rows and columns. It is powerful for creating complex page layouts. Grid provides better control for full-page responsive designs.

16. Difference between Flexbox and Grid?

Answer

Flexbox is mainly designed for one-dimensional layouts, either row or column. CSS Grid is designed for two-dimensional layouts involving both rows and columns. Flexbox is better for smaller layouts, while Grid is ideal for page-level layouts.

17. What is position property?

Answer

The position property controls how elements are positioned in the document. Common values include static, relative, absolute, fixed, and sticky. It is widely used for overlays, menus, and layout adjustments.

18. Difference between relative and absolute?

Answer

Relative positioning moves an element relative to its normal position. Absolute positioning removes the element from normal flow and positions it relative to the nearest positioned parent. Absolute elements can overlap other elements.

19. What is z-index?

Answer

z-index controls the stacking order of overlapping elements. Higher z-index values appear above lower values. It only works on positioned elements.

20. What is media query?

Answer

Media queries are used to create responsive web designs for different screen sizes and devices. They allow developers to apply different styles based on width, height, or device type. Media queries are important for mobile-friendly applications.

JavaScript Interview Questions

21. What is JavaScript?

Answer

JavaScript is a scripting language used to make web pages interactive and dynamic. It is used for DOM manipulation, event handling, API communication, and frontend logic. Modern web applications heavily depend on JavaScript.

22. Difference between var, let, and const?

Answer

var has function scope and can be redeclared. let has block scope and allows reassignment but not redeclaration in the same scope. const also has block scope but does not allow reassignment after initialization.

23. What is hoisting?

Answer

Hoisting is JavaScript behavior where variable and function declarations are moved to the top of their scope during execution. Function declarations are fully hoisted, while variables declared with var are initialized as undefined.

24. What is closure?

Answer

A closure is a feature where an inner function can access variables from its outer function even after the outer function has completed execution. Closures are commonly used for data privacy and callbacks.

25. What is callback function?

Answer

A callback function is a function passed as an argument to another function and executed later. Callbacks are commonly used for asynchronous operations such as API calls and event handling.

26. What is promise?

Answer

A Promise is an object used to handle asynchronous operations in JavaScript. It represents eventual success or failure of an operation. Promises have three states: pending, fulfilled, and rejected.

27. What is async/await?

Answer

async/await is a modern way to handle asynchronous code in JavaScript. async makes a function return a Promise, and await pauses execution until the Promise resolves. It makes asynchronous code cleaner and easier to read.

28. Difference between == and ===?

Answer

== compares only values and performs type conversion if needed. === compares both value and type without type conversion. Using === is recommended for safer and predictable comparisons.

29. What is event bubbling?

Answer

Event bubbling is the process where an event starts from the target element and propagates upward through parent elements. It is the default event flow in JavaScript. Bubbling helps in event delegation.

30. What is event delegation?

Answer

Event delegation is a technique where a parent element handles events for its child elements. It improves performance by reducing the number of event listeners. It is useful for dynamically created elements.

31. What is array map()?

Answer

map() is an array method used to transform each element and return a new array. It does not modify the original array. map() is commonly used for UI rendering and data transformation.

32. What is filter()?

Answer

filter() is an array method used to return elements that satisfy a condition. It creates a new array without modifying the original array. It is commonly used for search and filtering operations.

33. What is reduce()?

Answer

reduce() is an array method used to accumulate values into a single result. It is commonly used for calculating totals, grouping data, and transformations. reduce() takes an accumulator and current value.

34. What is destructuring?

Answer

Destructuring is a JavaScript feature used to extract values from arrays or objects into variables. It improves readability and reduces repetitive code.

35. What is spread operator?

Answer

The spread operator (...) expands arrays or objects into individual elements. It is commonly used for copying arrays, merging objects, and passing function arguments.

TypeScript Interview Questions

36. What is TypeScript?

Answer

TypeScript is a superset of JavaScript developed by Microsoft. It adds static typing, interfaces, classes, and advanced tooling support. Angular uses TypeScript because it improves scalability and maintainability.

37. What are data types in TypeScript?

Answer

TypeScript provides data types such as string, number, boolean, any, unknown, void, array, and object. Data types improve type safety and reduce runtime errors.

38. What is interface?

Answer

An interface defines the structure of an object in TypeScript. It specifies required properties and their data types. Interfaces improve readability and enforce type safety.

39. What is type alias?

Answer

A type alias creates a custom reusable type in TypeScript. It can represent primitive types, objects, unions, and function signatures. Type aliases improve code organization.

40. Difference between interface and type?

Answer

Interfaces are mainly used for object structures and support extension. Types are more flexible and can represent unions, primitives, and tuples. Both improve type safety in TypeScript.

41. What is enum?

Answer

Enum is a TypeScript feature used to define named constant values. It improves code readability and avoids magic values in applications.

42. What is optional property?

Answer

Optional properties are properties that may or may not exist in an object. They are defined using the ? symbol. Optional properties provide flexibility in object structures.

Example:

```
name?: string
```

43. What is readonly?

Answer

readonly is a TypeScript keyword used to prevent modification of properties after initialization. It improves immutability and protects object state.

44. What is generics?

Answer

Generics allow developers to create reusable components, functions, and classes while maintaining type safety. They help avoid code duplication and support dynamic typing.

45. What is access modifier?

Answer

Access modifiers control visibility of class members in TypeScript. Common access modifiers are public, private, and protected. They support encapsulation and secure code structure.

46. What is inheritance?

Answer

Inheritance allows one class to acquire properties and methods from another class. It improves code reusability and supports object-oriented programming principles.

47. What is polymorphism?

Answer

Polymorphism allows methods or objects to behave differently based on context. It is an important object-oriented programming concept used for flexibility and extensibility.

48. What is abstraction?

Answer

Abstraction hides implementation details and exposes only essential functionality. It simplifies complex systems and improves maintainability.

49. What is encapsulation?

Answer

Encapsulation is the process of binding data and methods together inside a class while restricting direct access to internal details. It improves security and maintainability.

50. Difference between any and unknown?

Answer

any disables type checking completely and allows any operation. unknown is safer because values must be type-checked before use. unknown is preferred when the type is uncertain.